

Activity 13 Instruction Manual

Activity 13 | Topic 02 | C013 | Version v1.0

Purpose

When delivery goes wrong, the visible problem is rarely the real one. You will run three complementary root-cause techniques on the same problem with GenAI's help and compare what each reveals.

Files used

- activity-guide.md and activity-guide.pdf - brief and source instructions
- prompts.pdf - reusable prompt and verification checklist
- mock-data.xlsx - safe fictional project data for practice

Expected output

A fishbone diagram, a five-whys chain and a Pareto analysis of the same delivery problem.

Before you begin

1. Form the required group and assign one person to capture evidence.
2. Open mock-data.xlsx and review the title, note and column headings.
3. Open prompts.pdf beside your chosen GenAI tool. Do not upload confidential workplace data.

Detailed procedure

1. In your group, choose one real delivery problem - recurring missed deadlines, rework, defects escaping to production.
2. Open 'Fishbone Diagram Template.pptx'. Prompt GenAI: 'What are the possible root causes of recurring project deadline misses in IT infrastructure upgrades?' - substituting your problem.
3. Sort the suggested causes into the fishbone categories (People, Process, Technology, Environment, Materials, Measurement). Reclassify anything the AI put in the wrong bone.
4. Add the causes GenAI could not know - the ones specific to your organisation's politics, tooling or history. These are usually the real ones.
5. Now run Five Whys on the single most likely cause. Prompt: 'Apply the Five Whys method to [problem] to uncover the root cause and identify potential solutions.'
6. Critique the chain. GenAI tends to produce a plausible but shallow chain that stops at 'insufficient planning'. Push each 'why' until you reach something you could actually change.
7. Open 'Pareto Principle Template.xlsx'. Prompt GenAI to generate a frequency count of defect or delay categories, enter them, and identify the vital few causing roughly 80% of the impact.
8. Compare the three techniques: which surfaced the actionable cause? Share your conclusion with the class.

Use the supplied mock data

Copy only the rows needed for the current step and tell the model the data is fictional. If the model changes an identifier or number, reject the change and rerun with an explicit preserve-values instruction.

Quality control

- You have all three artifacts for one problem, you added at least two organisation-specific causes the AI missed, and your five-whys chain ends at something you can change.
- Compare the first AI draft with the final human-corrected artifact and retain both as evidence.
- Label assumptions, decisions, owners and dates so another project manager can audit the result.

Troubleshooting

If the response is generic, add the project objective, constraints, audience and required columns. If arithmetic is inconsistent, request formulas and substitutions, then calculate independently. If a source cannot be opened, remove the claim or replace it with verified evidence.

Submission evidence

Submit the completed project artifact, the final prompt, a short critique of the AI output, and any screenshot or exported file required by the activity.